AI UPCHAR

Clinical Question & Reasoning Library — Master Specification

Phases 1–4 • Adaptive Clinical Interview • Next-Best-Question Engine • Doctor Decision Support

0. Purpose

This document is the master specification for AI UPCHAR's clinical interview intelligence. It is intended as an implementation/update brief for an AI/software team. The system must not behave like a fixed questionnaire. It must understand the patient's complaint, identify the relevant clinical pathway, continuously check safety, use prior history and uploaded documents, and select the most useful next question.

Clinical content must be reviewed, versioned and approved by qualified clinicians before production use. This document defines product architecture and question-design principles; it is not a substitute for medical guidelines or clinician governance.

1. Evidence & Design Basis

- WHO/ICRC Basic Emergency Care teaches a systematic initial approach to acute illness and injury, including ABCDE and SAMPLE history, with emphasis on time-sensitive conditions. □cite□turn0search12□turn0search39□
- WHO mhGAP uses condition-specific algorithms and structured clinical assessment for priority mental, neurological and substance-use conditions, supporting the principle of pathway-specific assessment rather than one generic questionnaire. □cite□turn0search1□
- Acute diarrhea assessment commonly considers onset, frequency, duration, stool character, fever, vomiting, urine output, abdominal pain, exposures, medications, travel and risk factors; testing depends on severity and clinical context. □cite□turn0search13□

External medical sources should be treated as evidence inputs with source, date/version and jurisdiction recorded. The system must not silently convert web search results into medical rules.

2. Four-Phase Roadmap

Phase 1 — High-Frequency Primary-Care Library

Build the core symptom/complaint pathways used most often in general OPD and tele-intake. Each pathway gets questions, branches, red flags, context requirements, document types, investigation considerations and report mappings.

Phase 2 — Specialty Expansion

Expand into cardiology, respiratory, gastroenterology, neurology, endocrinology, renal/urology, dermatology, ENT/eye, musculoskeletal, women's health, pediatrics and mental health.

Phase 3 — Rare & Specialist Pathways

Add lower-frequency, complex and specialty-specific pathways. The system should prefer specialist escalation/review when confidence or coverage is insufficient rather than inventing a pathway.

Phase 4 — Evidence, Governance & Continuous Learning

Add guideline/version management, clinical review, source provenance, audit trails, pathway performance monitoring, safety evaluation, question-quality feedback and controlled updates.

3. Core Intelligence Architecture

Patient language → symptom/complaint extraction → clinical concept normalization → pathway selection → context assembly → safety screen → candidate-question generation → question ranking → next-best-question → answer interpretation → context update → repeat.

- Patient language must be accepted in natural Hindi/Hinglish/English and mixed language.
- AI should recognize synonyms and colloquial descriptions.
- Do not require the patient to know a disease name.
- Use age, sex, pregnancy relevance, known history, medicines, allergies and previous visits when relevant.
- Never ask a question merely because it exists in a template.
- Stop or escalate normal questioning when a safety rule requires prompt clinical attention.

4. Clinical Pathway Object

Every pathway should be stored as structured data, not only as prose.

pathway_id, display_name, synonyms, specialty, age applicability, sex/pregnancy considerations, opening questions, core question bank, branch rules, red flags, risk factors, relevant history, medication questions, allergy questions, exposure questions, document types, image-upload needs, differential considerations, investigation considerations, escalation rules, report template, evidence sources, version, clinical reviewer, last reviewed date

5. Question Object

Question record: question_id, text, category, answer_type, trigger, priority, safety_weight, information_gain, branch_conditions, skip_conditions, required_context, source, version

Answer types: free text, single choice, multiple choice, numeric, date/time, severity scale, yes/no, measurement, file upload, image upload.

6. Next-Best-Question Engine

For every turn, generate candidate questions and rank them. Conceptual priority = safety relevance + complaint relevance + information gain + context relevance + change-from-history + document relevance − redundancy − irrelevance.

- Safety questions can override ordinary ranking.
- Questions already answered reliably should not be repeated.
- Known historical information should be confirmed only when clinically relevant or potentially changed.
- Prefer one high-value question over several low-value questions.
- After a document is processed, ask only questions that the document leaves unresolved.
- If no useful question remains, stop intake rather than artificially extending the chat.

7. Red-Flag / Escalation Engine

- Maintain pathway-specific red flags.
- Maintain general acute-illness safety checks.
- Use age-specific and pregnancy-specific safety rules.
- Escalation must be explainable: show the signal that triggered it.
- AI can flag urgency and recommend clinician review; final disposition remains with the clinician.
- Emergency-style pathways should be clinically validated before deployment.

8. Documents as Clinical Context

- Prescription → identify previous medicines/duration/instructions where reliably extractable.

- Lab report → extract test names, dates, values, units and reference ranges where available.
- Imaging report → extract report text, impression and date.
- Discharge summary → extract admission reason, diagnoses recorded by treating team, procedures, medicines and follow-up instructions.
- Referral letter → extract referring concern and requested evaluation.
- Document findings must retain provenance and confidence.
- AI must distinguish extracted fact from interpretation.

9. Appointment & Longitudinal Context

Each appointment has a separate immutable AI chat transcript. Patient profile contains all appointment histories. New appointments load a concise longitudinal context: last visit reason, key changes, doctor-approved treatment/prescription, tests advised/results, follow-up date and relevant prior chat/document summaries.

Raw transcript remains viewable; the AI uses structured summaries to avoid unnecessarily reprocessing every historical message.

10. Disease / Complaint Library — Phase 1

Domain	Initial pathway set
General symptoms	Fever; fatigue/weakness; weight loss; generalized pain; dizziness; fainting
Gastrointestinal	Acute diarrhea; persistent diarrhea; abdominal pain; vomiting; constipation; GERD/heartburn; dyspepsia/gastritis-like symptoms; jaundice
Respiratory	Cough; sore throat; common cold/URI; breathlessness; wheeze; acute bronchitis-like presentation; pneumonia-like presentation
Cardiovascular	Chest pain; palpitations; high blood pressure follow-up; leg swelling
Neurological	Headache; migraine-like presentation; vertigo/dizziness; numbness/tingling; weakness; seizure/fainting history
Metabolic/endocrine	Diabetes/high glucose; low glucose symptoms; thyroid symptoms; abnormal weight change
Urinary/renal	Burning urination; frequency/urgency; UTI-like symptoms; flank pain/kidney-stone-like presentation; blood in urine
Musculoskeletal	Back pain; neck pain; knee pain; shoulder pain; joint pain; muscle pain
Dermatology	Rash; itching; acne; fungal-like rash; dermatitis/eczema-like presentation; skin infection-like presentation; hair loss
ENT/eye	Ear pain; hearing change; red eye; eye pain; blurred vision; sinus symptoms; sore throat
Women's health	Menstrual irregularity; heavy bleeding; pelvic pain; vaginal discharge; pregnancy-related symptoms; menopause symptoms
Pediatrics	Child fever; child cough/breathing difficulty; child diarrhea; child vomiting; feeding problem; rash; growth concern
Mental health	Anxiety symptoms; low mood/depression symptoms; panic symptoms; sleep problem; memory/cognitive concern

11. Phase 1 Pathway Question Blueprint

Every Phase 1 pathway should be authored using this sequence, then customized:

- 1. Patient's own complaint
- 2. Onset and duration
- 3. Frequency/pattern
- 4. Location/character where relevant
- 5. Severity/functional impact
- 6. Associated symptoms
- 7. Red flags

- 8. Relevant medical history
- 9. Medicines/allergies
- 10. Previous episodes/treatment
- 11. Exposure/risk factors
- 12. Relevant family/social history
- 13. Relevant documents
- 14. What changed since last visit
- 15. Missing information
- 16. Investigation considerations
- 17. Clinician-review considerations

12. Example Pathway — Persistent Diarrhea

Trigger examples: loose motion/diarrhea for a prolonged period. The engine should distinguish acute vs persistent/chronic duration and branch accordingly.

- How long has it been happening?
- How many bowel movements per day?
- What is the stool character?
- Any blood, black stool or mucus?
- Fever/chills?
- Abdominal pain/cramping and location?
- Vomiting/nausea?
- Reduced urine, thirst, dry mouth, dizziness or marked weakness?
- Unintentional weight loss?
- Recent travel, outside food or unsafe water?
- Sick contact or outbreak exposure?
- Recent antibiotics or new medicines?
- Previous gastrointestinal disease or surgery?
- Immune compromise or major comorbidities?
- Previous tests or treatment?
- Upload previous prescription/report if available.

Testing should be presented as clinician considerations based on findings, not automatic patient orders. For example, severe/inflammatory features and longer duration can change the need for further evaluation. □cite□turn0search13□

13. Example Pathway — Chest Pain

- Onset/time pattern
- Current pain now?
- Location
- Pressure/tightness/burning/sharp quality
- Radiation to arm/jaw/back
- Exertional relationship
- Breathlessness
- Sweating/nausea
- Fainting/dizziness
- Known heart disease
- Blood pressure/diabetes/cholesterol history
- Smoking/tobacco
- Previous episodes/investigations
- Medication history
- Immediate red-flag escalation if indicated

This pathway requires clinician-reviewed emergency rules before production.

14. Example Pathway — Headache

- First or recurrent?
- Sudden vs gradual onset
- Time course/frequency
- Location
- Severity
- Duration of each episode
- Light/sound sensitivity
- Nausea/vomiting
- Vision change
- Weakness/numbness
- Speech/confusion
- Fever/neck stiffness
- Recent head injury
- Pregnancy/postpartum relevance where applicable
- Previous migraine/headache history
- Current medicines and prior treatment

15. Example Pathway — Cough / Breathing

- Duration
- Dry vs productive
- Sputum
- Blood in sputum
- Fever
- Breathlessness
- Wheeze
- Chest pain
- Night symptoms
- Smoking/tobacco
- Allergy/asthma history
- TB exposure/relevant contact
- Weight loss/night sweats where relevant
- Previous respiratory disease
- Current medicines
- Upload previous reports/imaging if available

Acute breathing complaints should also be assessed through validated safety/acuity pathways where appropriate; WHO BEC uses ABCDE and SAMPLE for systematic acute assessment. □cite□turn0search12□

16. Example Pathway — Diabetes / High Glucose

- Known diabetes or new finding?
- Recent glucose value and whether fasting/post-meal/random
- How long elevated
- Current medicines
- Adherence/changes
- Excess thirst
- Frequent urination
- Weight change

- Blurred vision
- Weakness
- Recent infection/illness
- Recent laboratory reports
- Previous HbA1c where available
- Previous treatment changes
- Hypoglycemia episodes

17. Example Pathway — Urinary Symptoms

- Onset
- Burning
- Frequency
- Urgency
- Lower abdominal pain
- Fever/chills
- Flank/back pain
- Blood in urine
- Urine appearance/odor
- Previous UTI
- Recent antibiotics
- Pregnancy relevance
- Kidney/urinary history
- Previous urine tests

18. Example Pathway — Back Pain

- Onset
- Injury/lifting
- Location
- Radiation to leg
- Numbness/tingling
- Weakness
- Walking/gait change
- Bladder/bowel changes
- Fever
- Unexplained weight loss
- Cancer/infection risk history
- Previous spine disease
- Previous imaging
- Medicines/treatment

19. Example Pathway — Skin Rash

- Onset
- Location/distribution
- Spread
- Itching
- Pain/burning
- Blisters
- Fever/systemic symptoms
- New medicine
- New food/product/cosmetic

- Contact exposure
- Others affected
- Previous episodes
- Photo upload
- Previous treatment

20. Example Pathway — Mental Health

- Main concern in patient's own words
- Onset/duration
- Impact on daily life
- Sleep
- Appetite/weight
- Energy/concentration
- Anxiety/panic symptoms
- Mood changes
- Substance use where relevant
- Previous mental-health treatment
- Current medicines
- Psychosocial stressors
- Safety/self-harm screening when indicated
- Support system

Mental-health pathways should follow validated, clinician-reviewed frameworks. WHO mhGAP uses condition-specific algorithms and follow-up-oriented clinical assessment. □cite□turn0search1□

21. Phase 2 — Specialty Expansion

- Cardiology: chest pain, palpitations, hypertension, edema, heart-failure symptoms.
- Pulmonology: asthma/COPD-like symptoms, persistent cough, pneumonia-like symptoms, TB evaluation pathway.
- Gastroenterology: GERD, dyspepsia, IBS-like symptoms, inflammatory bowel disease considerations, chronic liver symptoms, gallbladder symptoms.
- Neurology: migraine, seizure, syncope, neuropathy, movement/cognitive concerns.
- Endocrinology: diabetes follow-up, thyroid, obesity/weight change, metabolic abnormalities.
- Nephrology/Urology: CKD follow-up, stones, hematuria, urinary symptoms.
- Orthopedics/Rheumatology: arthritis, inflammatory joint symptoms, musculoskeletal injury.
- Dermatology: acne, eczema, fungal disease, psoriasis-like symptoms, skin infections.
- ENT/Ophthalmology: sinus, hearing, eye pain/redness, visual change.
- Gynecology/Obstetrics: menstrual, pelvic, pregnancy and postpartum pathways.
- Pediatrics: age-specific fever, respiratory, GI, dehydration, growth/development pathways.
- Mental/Neurological/Substance use: condition-specific assessment pathways using validated clinical guidance.

22. Phase 3 — Rare & Specialist Pathways

For rare/complex presentations, the engine should first identify the dominant symptom/problem and route to a specialist pathway or clinician review. It must not fabricate a rare-disease diagnosis.

- Rare disease aliases/synonyms
- Complex multi-system symptoms
- Post-operative complications
- Immunocompromised patient pathways
- Complex medication adverse-event pathways
- Cancer-related symptom pathways

- Specialist referral preparation
- Complex chronic disease follow-up
- Multi-morbidity context

23. Phase 4 — Governance & Continuous Updating

- Every pathway has version, source, jurisdiction, publication/update date and clinical reviewer.
- Guideline changes create a new pathway version rather than silently editing history.
- Old patient records retain the version used at the time.
- AI outputs retain source/provenance.
- Clinicians can flag unsafe, irrelevant or missing questions.
- Question performance can be measured: skipped rate, answer quality, downstream usefulness, escalation capture and clinician feedback.
- Any change to red-flag or treatment-related logic requires clinical review and controlled release.
- External web retrieval must not directly overwrite production clinical rules.
- Maintain an audit trail of pathway and model changes.

24. AI Report Structure

- Patient & appointment
- Chief complaint
- History of present complaint
- Relevant positives
- Relevant negatives
- Past history
- Medication/allergy history
- Previous treatment
- Documents and extracted findings
- Previous visit comparison
- Safety/red flags
- Differential/clinical considerations
- Missing information
- Potential investigations to consider
- Urgency considerations
- Questions for clinician to clarify
- Evidence/source context
- Doctor decision

The report must label AI-generated considerations clearly. It must not present an AI hypothesis as a confirmed diagnosis.

25. What AI May Suggest vs What Doctor Must Decide

AI may prepare	Doctor decides
Relevant questions	Final diagnosis
Structured history	Final treatment
Potential clinical considerations	Prescription
Safety/urgency flags	Investigation orders
Potential investigations to consider	Admission/disposition
Document summaries	Surgery/procedure decision
Previous-visit comparison	Final patient instructions

26. Implementation Rules for the AI

- Never ask unrelated generic questions when a complaint-specific pathway exists.

- Never repeat a question whose answer is already known unless confirmation is clinically useful.
- Never assume a patient understands medical terminology.
- Use simple language and allow the patient to answer naturally.
- Ask one primary question at a time.
- Allow “I don't know” and “not sure”.
- Allow document upload at relevant points.
- Use uploaded documents to reduce redundant questioning.
- Prefer stopping once enough context is collected rather than maximizing chat length.
- Safety rules take priority over convenience.
- Keep AI inference separate from extracted facts.
- Keep patient-reported information separate from doctor-confirmed diagnosis.
- Keep AI suggestions separate from doctor-approved decisions.
- Maintain provenance and timestamps.

27. Complete Four-Phase Delivery Plan

1. Phase 1: Build the common symptom/complaint engine and high-frequency pathways; validate with clinicians.
2. Phase 2: Add specialty pathways and age/sex/pregnancy-specific branches.
3. Phase 3: Add rare/complex pathways and specialist escalation.
4. Phase 4: Establish evidence/version governance, audit, clinical review, safety testing and continuous improvement.
5. Integrate the engine with Patient 360, appointment-specific chat history, document intelligence and AI Pre-Visit Brief.
6. Only after validation expose the resulting preparation in the Doctor Workspace UI.

28. Final Instruction to the Implementing AI

Do not interpret this document as a request to create one enormous static questionnaire. Implement an adaptive clinical interview system. The AI must understand what the patient has already said, what is already known from the patient's history, what documents have been provided, which clinical pathway is relevant, which safety questions matter, and what single next question will add the most useful information. The goal is to prepare a structured, evidence-aware, clinician-reviewable patient brief—not to replace the clinician's final medical judgment.